

SURYANSH KUMAR PATHAK

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Experience

Technical Lead

Apr 2024 – Present

Microsoft (via HCLTech)

Redmond, WA

- Architected Microsoft Copilot for M365 Admin Center, a GenAI-powered assistant **serving 500K+ enterprise administrators**, integrating Azure OpenAI, MCP servers, grounded retrieval, Bot Framework, and LUIS to **automate 45% of Tier-1 support requests**, **save 10K+ support hours annually**, and **improve CSAT by 25%**.
- Designed and implemented scalable distributed systems and high-throughput RESTful microservices **supporting millions of daily transactions** and **2M+ requests/day**, achieving **99.95% availability**, **sub-150ms P95 latency**, **30% higher throughput**, **2× scalability during peak traffic**, and **45% lower API response times**.
- Built enterprise-grade AI evaluation and quality assurance frameworks using DeepEval/SEVAL, including golden datasets, regression detection, signal-quality metrics, and failure classification, **improving Copilot response quality by 18%** and **reducing post-release defects by 35%**.
- Led development of Microsoft Fraud Protection backend services powering real-time fraud detection, risk evaluation, rule-based decisioning, and alerting across Microsoft commerce platforms, **reducing fraud detection latency by 40%**, increasing protection coverage, and **lowering manual investigation effort by 35%**.
- Strengthened platform security, reliability, and governance by implementing RBAC enforcement, privilege-escalation guardrails, and modernizing legacy services, eliminating authorization risks identified during security reviews while **improving system reliability to 99.9% availability**.

Software Engineer

Oct 2023 – Apr 2024

Serigor

Baltimore, MD

- Designed event-driven microservice architecture using Python, RabbitMQ, and REST APIs, processing **50K+ events/day** with **99.9% reliability** and **reducing processing time from 5s to 200ms**.
- Optimized PostgreSQL and AWS Aurora database handling **10M+ student records** through schema redesign and query tuning, reducing **P99 latency by 65%** and improving dashboard load times by 3x.
- Established scalable CI/CD pipeline with Selenium Grid and JMeter, executing **5K+ test cases nightly**, **reducing regression testing time by 70%** (from 8 hours to 2.5 hours).

Software Engineer

Jul 2023 – Oct 2023

Softworld Technologies

Des Moines, IA

- Developed and deployed RESTful microservices using Java, Spring Boot, and AWS, supporting internal platform capabilities and API integrations.
- Assisted in building and maintaining CI/CD pipelines using Jenkins, Docker, and GitHub Actions, helping **reduce build and deployment turnaround times by approximately 25%**.
- Created 50+ unit and integration tests** using JUnit and Mockito, improving code coverage and contributing to more stable releases through early defect detection.

Technical Skills

Languages: C++, C#, Python, Java, JavaScript/TypeScript, SQL, HTML/CSS

Frameworks: ASP.NET Core, Spring Boot, Node.js, Express, REST APIs, GraphQL, Bot Framework

Cloud / Distributed Systems: Azure Service Fabric, Azure OpenAI, Azure Functions, AWS Aurora, AWS Lambda, Kafka, RabbitMQ, Redis

Databases: PostgreSQL, MongoDB, MS SQL Server, AWS Aurora, Redis

DevOps / Observability: Docker, Kubernetes, Jenkins, Git, Azure DevOps, Postman, JMeter, Selenium Grid

AI / LLM System: Azure OpenAI, RAG, MCP servers, LangChain, LangGraph DeepEval, SEVAL, LUIS

CS Fundamentals: Data Structures & Algorithms, System Design, OOP, Operating Systems, DBMS, Distributed Systems

Education

University of Massachusetts, Lowell

Sep 2021 – May 2023

Master of Science in Computer Science

Lowell, MA

Madan Mohan Malaviya University of Technology

July 2017 – May 2021

Bachelor of Technology in Computer Science & Engineering

Gorakhpur, India